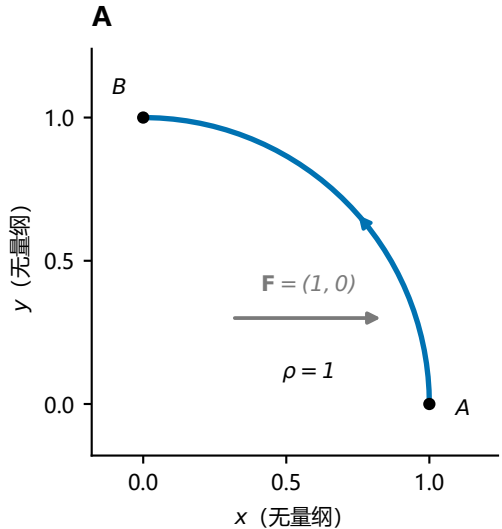
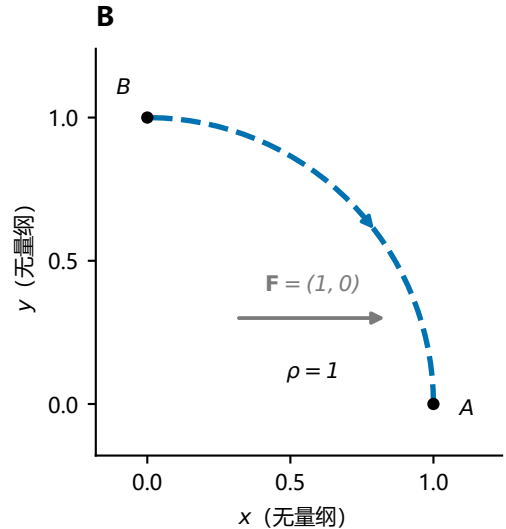


$C : A \to B$

$\int_C ds = \pi/2, \quad \int_C dx = -1$



$-C : B \to A$

$\int_{-C} ds = \pi/2, \quad \int_{-C} dx = 1$